Adjusting for covariates

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http://apps.who.int/gho/athena/data/GHO/WHOSIS_000008.csv?profile=text&filter=COUNTRY:;SEX:

WE CAN END POVERTY 2015 MILLENNIUM DEVELOPMENT GOALS

UNITED NATIONS SUMMIT
20-22 September 2010, New York
High-level Plenary Meeting of the General Assembly

WE CAN

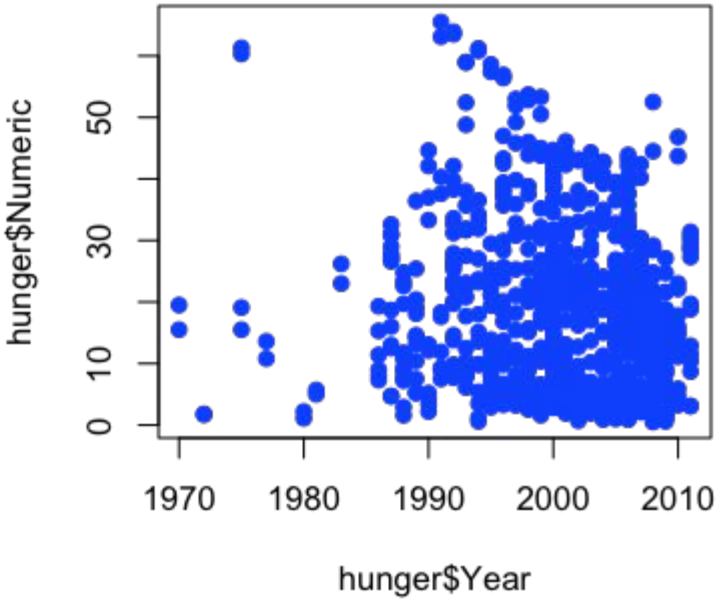
FACT SHEET

GOAL 1
Eradicate Extreme Poverty and Hunger

TARGETS

1. Halve, between 1990 and 2015, the proportion of people whose income is less than $1 a day
2. Achieve full and productive employment and decent work for all, including women and young people
3. Halve, between 1990 and 2015, the proportion of people who suffer from hunger

https://github.com/jtleek/dataanalysis/blob/master/week4/006multipleVariables/



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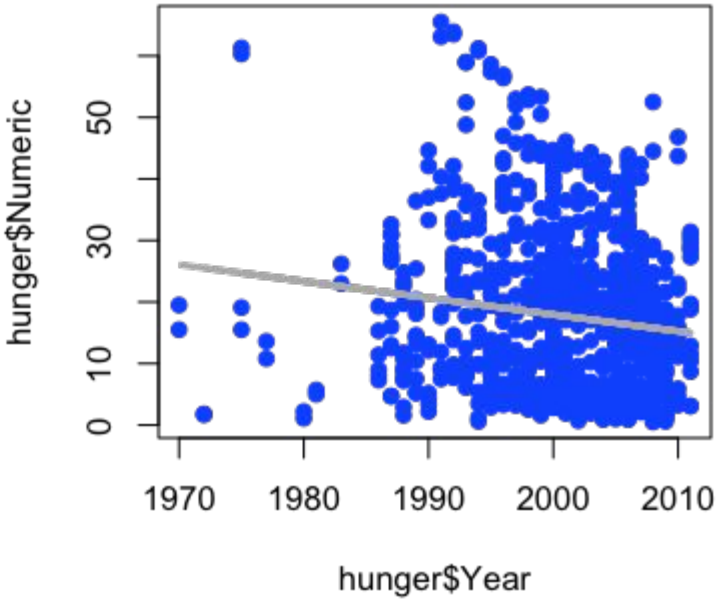
$$Hu_i = b_0 + b_1 Y_i + e_i$$



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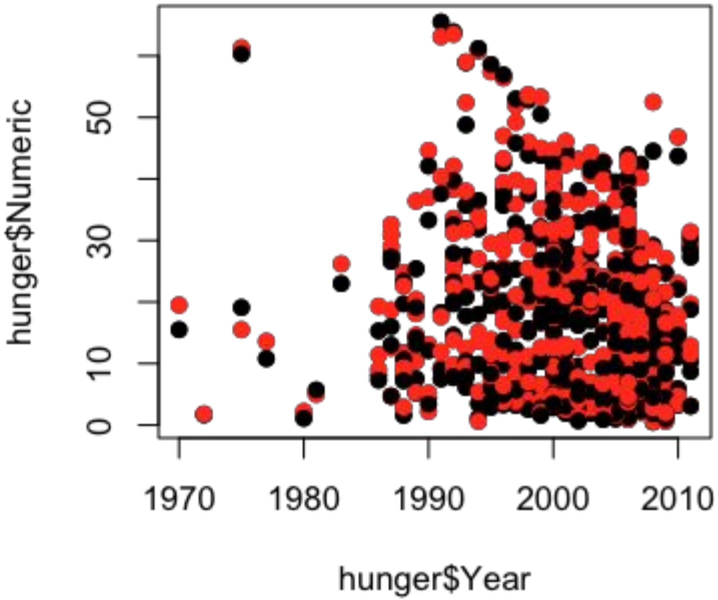
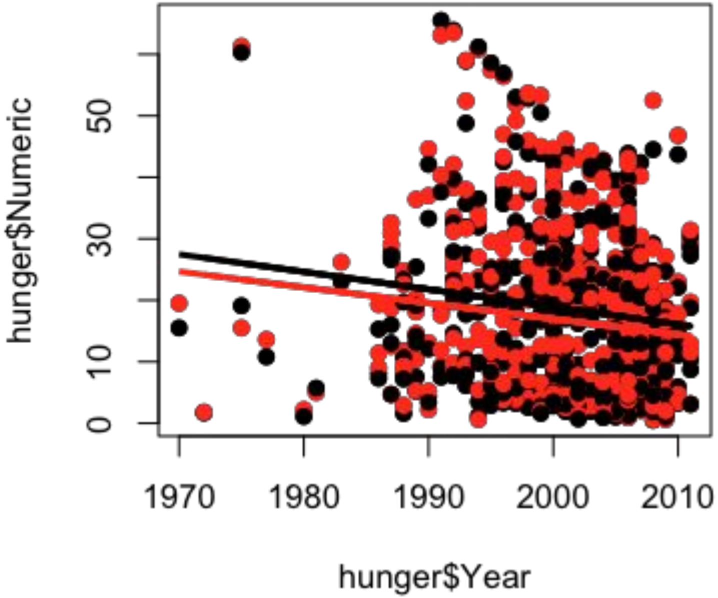


$$Hu_i = b_0 + b_1 Y_i + b_2 F_i + e_i$$

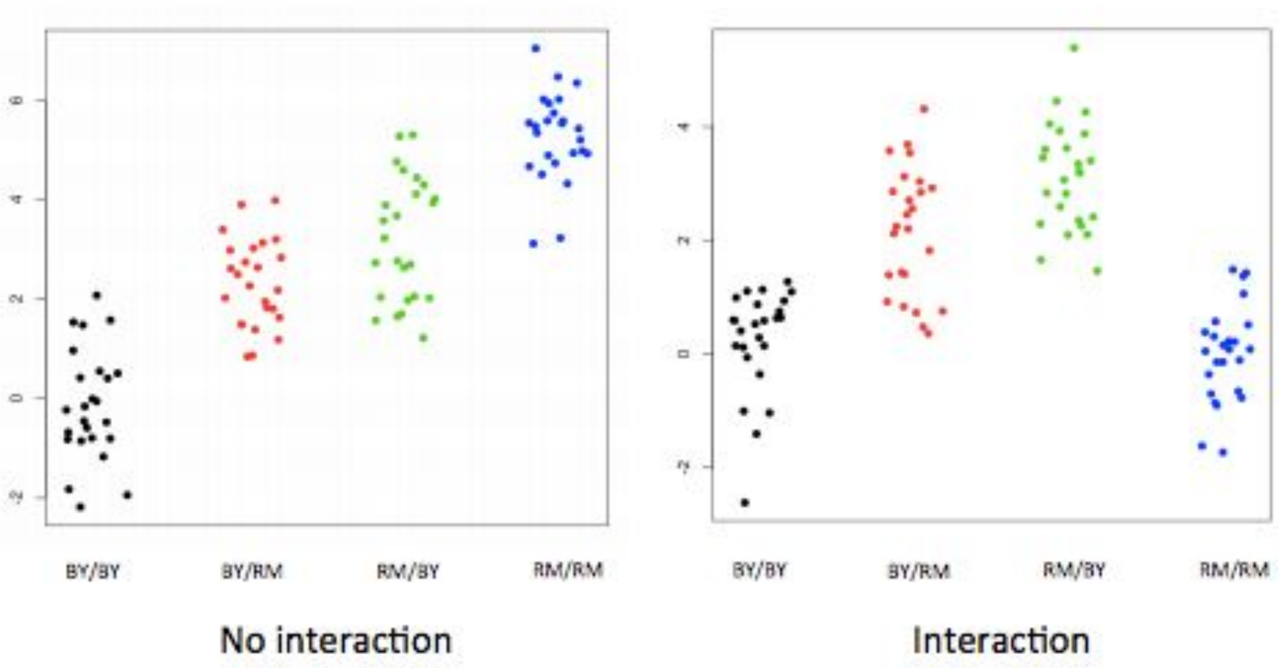


b_0 - percent hungry at year zero for males

$b_0 + b_2$ - percent hungry at year zero for females

b_1 - change in percent hungry (for either males or females) in one year

e_i - everything we didn't measure



Expression = Baseline + RM Effect + BY Effect + (RM Effect * BY Effect) + Noise

Notes and further reading

- Linear models is a whole class (no joke): https://www.coursera.org/course/regmods
- Basic thing to keep in mind is how many levels do you want to fit? What makes sense biologically?
- Great additional notes in Chapter 2 here: http://genomicsclass.github.io/book/